

## **\*\*Data Engineer - VNG Corporation\*\***

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 [dataspi.github.io/DataSpi](https://dataspi.github.io/DataSpi)

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*Data/Analytics Engineer with 3+ years of experience turning raw operational data into trusted, well-modeled datasets for analytics. Strong in `Python`, `SQL`, `PostgreSQL`, `BigQuery`, `dbt`, and `AWS Glue`, with hands-on delivery of dimensional data models, tested ELT pipelines, and a self-service semantic layer ( `Malloy` ). Proven ability to bridge engineering and business stakeholders — having worked as both a BI/HR analyst and a data engineer — to deliver data products people actually trust and use, and to mentor non-technical teammates into full ownership of the systems I build.*

## Highlighted Projects

More details on my project, visit [Projects](#)

### 1. Real Estate Analytics Platform

Sep 2025 – Present

Built and now operate an end-to-end analytics engineering pipeline for Vietnam's real estate market (~100K+ listings across HCMC & Hanoi), following a medallion architecture on `BigQuery`: async scrapers land raw listings, `dbt` staging models deduplicate and clean into a silver layer, and `dbt` mart models build a gold layer surfaced through a `Malloy` semantic layer for self-service metrics. Data quality is enforced with `dbt` schema tests plus custom singular tests (e.g. negative price/area, price-per-m<sup>2</sup> outliers). The pipeline runs on a scheduled, fail-fast orchestrator with logging, and results are published via an interactive `Looker Studio` dashboard and detailed analysis reports.

**Tech stack:** `Python`, `BigQuery`, `dbt`, `Malloy`, `curl_cffi`, `BeautifulSoup`, `Google Looker Studio` [GitHub](#), [Publication](#)

### 2. CI/CD-Orchestrated Logistics Reporting Pipeline

Nov 2025 – Present

Built an event-driven ELT pipeline on `GitHub Actions` for two freight-forwarding clients (retail & food logistics), replacing manual daily reporting. Scheduled workflows scrape order/booking data into `BigQuery` bronze tables, then automatically trigger a chained workflow ( `workflow_run` ) that runs `dbt build` for silver/mart models, followed by a report workflow that pushes milestone and TEU reports to stakeholders — a full scrape-to-report pipeline with zero manual intervention. Engineered a reusable composite action for secrets/env provisioning across pipelines, plus failure-artifact capture ( `upload-artifact` ) and manual `workflow_dispatch` triggers for on-demand reruns and debugging.

**Tech stack:** `Python`, `GitHub Actions` (CI/CD, cron & event-driven orchestration), `dbt`, `BigQuery`, `Google Sheets` [API](#) [GitHub](#)

### 3. Building an Internal AI Assistant \*(Sep 2023 - Present)\*

Internal assistant for policy and knowledge lookup, letting employees retrieve company information in natural language.

`OpenAI API`, `Langchain`, `Pinecone` - [GitHub](#)

#### 4. Labor Market Analysis from Job Listing Sites \*(Aug 2023 - Present)\*

Automated job posting collection and analysis to track hiring trends in Vietnam's data job market. `Python`, `Selenium`, `BeautifulSoup` - [GitHub](#), [Publication](#)

#### 5. Employee Mental Health & Stress Survey Analytics \*(Aug 2023 - Sep 2023)\*

Analyzed employee and manager survey data to compare stress and mental health perceptions across cohorts. `Pandas`, `Google Forms` — [Project Link](#)

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## Work experience

### VNG Corporation - Data Engineer

Aug 2025- Now (1yr)

**Technical Core:** `Python`, `PostgreSQL`, `AWS Glue`, `Hasura`.

#### Key Projects & Impact:

- Data Platform & Migration: Designed and operated an `Odoo-to-ATS` data migration pipeline with reconciliation controls to maintain data integrity and prevent data loss at go-live.
- ETL/ELT Automation: Developed `Python` + `Boto3` automation for `AWS Glue` jobs, reducing manual operations and improving pipeline stability and run-time consistency.
- Managed `PostgreSQL` / `Hasura` workloads, including automated monthly partitioning, custom SQL functions, and scheduled maintenance ( `pg_cron` ) to keep query latency stable as data volume grew.
- Data Modeling & Consumption: Integrated `Malloy` semantic modeling into the `IPay` environment to improve self-service analytics and natural language querying accuracy.
- Monitoring & Documentation: Established technical documentation standards and implemented monitoring/alerting flows with operations teams to improve reliability and incident response.

### LEGO Manufacturing Vietnam - Data Analyst

Sep 2024 - Aug 2025 (1yr)

- Reverse-engineered `UKG` timesheet data to resolve a high-stakes overtime calculation conflict (legal 300h/year cap vs. LEGO's actual pay policy vs. shift-based break rules); presented the logic to C&B leadership and cross-functional DA teams (Packing, Moulding), and it was adopted as the standard OT calculation used factory-wide — ahead of a global system fix that remained pending over a year later.
- Mentored a non-technical successor to full ownership of the PPC HR `Power BI` reporting system through a 3-week hands-on training program and detailed documentation; the system ran uninterrupted for 6+ months post-handover, and I remained the trusted technical advisor for two successive report owners after leaving LEGO.
- Designed and delivered 3 targeted reporting solutions for the PPC department, covering: headcount & workforce planning, turnover, D&I, L&D, overtime, absenteeism, retention policy analysis to support strategic decision-making.
- Produced monthly executive summaries from `Power BI` dashboards for senior leadership.
- Partnered with BI Manufacturing on Factory Huddle and Visual Factory reports used in daily operations and leadership meetings.
- Delivered an end-to-end data flow for LEGO Vietnam's 3-month Active Summer program using `Google Forms API` + `Python` + `Power BI`.
- Supported development of a `Power App` solution for gym consent and reservations, enhancing process efficiency and user experience.

### ITL Corporation

HR Data Analyst

Aug 2023 - Sep 2024 (1yr 1m)

- Conducted comprehensive analyses and generated detailed reports using data from the HRMS database.
- Developed a comprehensive data pipeline spanning database, data warehouse, and reporting phases using `MSSQL` and `PostgreSQL`.
- Streamlined data using `SQL` and `Python`.
- Visualized and created reports using `Power BI`.
- Collected and analyzed job listings from job-listing sites using `BeautifulSoup`, `Selenium`, and `Pandas` to generate comprehensive reports on the recruiting market.
- Led comprehensive internal surveys to assess key HR metrics, including stress levels, employee satisfaction, and employee engagement.
- Collaborated with IT to develop an internal AI assistant leveraging the `OpenAI API` and `Langchain`.

## Human Resources Management Trainee

Feb 2022 - Aug 2023 (1yr 7m)

- Calculated monthly salary add-ons for the container fleet to ensure accurate and timely compensation adjustments.
- Provide support by answering questions related to company policies.
- Processed employee paperwork related to union matters and resignations.
- Planned employee costs for 2024.

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## Skills

### Technical Skills

#### Analytics Engineering

- Dimensional Modeling (Star Schema), Medallion Architecture (Bronze/Silver/Gold), Data Warehouse & Data Marts
- ETL/ELT Pipelines, `dbt` (models, schema & custom data quality tests), Semantic Modeling ( `Malloy` )

#### Data & Programming

- Advanced `SQL` ( `PostgreSQL`, `BigQuery`, `MSSQL` )
- `Python` ( `pandas`, `BeautifulSoup`, `Selenium`, `matplotlib`, `plotly` )

#### Data Platform & CI/CD

- `PostgreSQL`, `BigQuery`, `dbt`, `AWS Glue`, `Airflow`, `pg_cron`, `Git`, `Hasura`
- `GitHub Actions` (scheduled & event-driven workflow orchestration, dbt-in-CI, secrets management, reusable composite actions)

#### Business Analytics

- `Power BI`, `DAX`, KPI Development, Executive Reporting, Self-service Analytics

#### Collaboration

- Cross-functional stakeholder collaboration, technical documentation, mentoring & knowledge transfer, AI-assisted development ( `Cursor`, `VS Code` )
- Public writing on data careers & self-taught learning — [Threads](#) post reached 4K+ views, 347 likes, 54 reposts; longer-form companion piece on [Substack](#)

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## Soft Skills

### Languages

- Vietnamese: Native or bilingual proficiency
- English: Full professional proficiency (IELTS 7.0)

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## Education

### Bachelor Degree

- Human Resources Management - University of Economics HCMC - (*Sep 2018 - Dec 2021*)
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